Exercise Sheet

Using LLMs to Programmatically Extract and Curate Research Data

ResBaz NZ | Participant copy

Setup

1. Create folder: `data curation workshop resources 30 June 2026`
2. Download `plants_site_A.csv`, `plants_site_B.csv`, `plants_sites_AB_large.csv` from Zoom chat
3. Open Copilot at copilot.microsoft.com
4. Open Word — title: `Prompt log — data curation workshop — 30 June 2026`
5. Open data in Excel

Segment 1 — Intro + framing

Segment 2 — Block 1: Data cleaning

Research task completion framework

Use these steps for every exercise.

Step		What you do
1	**Identify task**	Understand what is required
2	**Identify potential issues**	Look at the data, run a prompt, note what's wrong in Word
3	**Copilot: fix**	Run a prompt to fix the issue
4	**Validate outputs**	Check the issue is fixed and the task is complete
5	**Log it in Word**	Note the prompt and what changed
6 *(if time allows)*	**Apply**	Use the same prompt on other columns of the same type

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2.0 Stage 1 — Data exploration

Drag and drop `plants_site_A.csv` into Copilot.

Prompt →

```
I am a researcher studying the effect of weed management on native plant
restoration across two sites in West Auckland. I want to understand whether
weeding alongside native planting (Site A) produces better plant outcomes
than native planting alone (Site B).

Attached is my data set (plants_site_A)

Tell me what you understand about this data — the columns, and what they seem
to measure.
```

Stage 2 — Numeric: `Height_cm`

Steps 1-2: Identify

Prompt →

```
Look at `Height_cm` only. Return a table with the mean, min, max, range.
```

Prompt →

```
Look at `Height_cm` only. Return a two-column table with the original column
(`Hieght_cm`), and a new column (`Height_cm_cleaned`) with potential outlier
values marked with "check".
```

Step 3: Fix

Prompt →

```
Modify value 340 to 34 and return the two-column table, do not include any notes
in the new cell, jsut "34".
```

Step 4: Validate — paste into `plants_site_A`; check first, last, changed rows

Step 5: Log — copy prompt to Word table `| column | prompt | summary of changes |`

Step 6: Apply

Prompt →

```
Provide me the same summary stats and outlier checks for `Num_Leaves` and
`Num_Flowers`.
```

Prompt →

```
Modify values and return four column table (original column + _cleaned):
`Num_Leaves`: 847 -> 87.
`Num_Flowers`: -3 -> 3. Keep 22 as 22.
```

2.3 Categorical: `Site`, `Location`, `Species`

Steps 1–2: Identify — Excel: Data → Filter, check `Species` dropdown

Step 3: Fix

Prompt →

```
Clean categorical columns `Site`, `Location`, `Quadrat`, `Species` and
`Flowering`.
I want them to be consistently spelt, all values lower-case, and values
separated by underscores "_".

Return pairs of columns for each variable: the original and a new column
`[column_name]_cleaned` for
each. In the cleaned column have the new, corrected value and use "check" if you
are unsure.
```

Step 3b (optional): Condition

Prompt →

```
For `Condition`, return the original column and a new column `Condition_cleaned`
with suggested categories representing a consistent five-point scale from poor
to excellent.
```

Step 5: Log — copy prompts to Word

2.4 Block 1 close

Segment 3 — Block 2: Feature extraction

3.1 Plant_Age_Months — guided

Prompt →

```
Look at `Date_Planted` and `Date_Measured`. These columns contain dates
in inconsistent formats — some DD/MM/YYYY, some D/M/YY, some DD-Mon
with no year, and possibly some US-format (M/D/YY).

Without standardising the formats first, calculate the age of each plant
in whole months at the time of measurement. Add this as a new column
called `Plant_Age_Months`. Return these three columns as a table.
Flag any row where you cannot confidently interpret the dates with "check".
```

Step 4: Validate — paste into spreadsheet; check row count, spot-check 3 rows, flag any negatives

Step 5: Log

3.2 Plant_ID — your turn

Task: Create a `Plant_ID` column — a unique identifier built from Site, Location, Quadrat, and species initials. Write your own prompt first. Use the one below if you're stuck.

Prompt → *(use this if you're stuck)*

```
Using `Site_cleaned`, `Location_cleaned`, `Quadrat_cleaned`, and
`Species_cleaned`,
create a new column `Plant_ID`. Format example: Sa_L1_Q1_LS (Site = A, Location
= 1, quadrat = 1, Leptospermum scoparium)
Return the only the new column.
```

Step 4: Validate — sort by `Plant_ID` in Excel; check all values are unique

Step 5: Log

3.3 Notes — your turn

Prompt →

```
I am about to analyse `Plant_Age_Months`, `Height_cm`, and `Condition` for these
plants. For each row, I want to know whether the `Notes` field gives any reason
this row's
measurements might be unreliable or need special handling.

Return two new columns: `Data_Quality_Flag`: Y or N; and `Data_Quality_Notes`: a
short reason for any Y.
```

Prompt →

```
Provide me a row-by-row thematic analysis of the `Notes` field, limit your
analysis of each comment into one of three, short distinct themes from the
perspective of a botanist that you will decide. Include the analysis as column
`Notes_themes`.
```

Step 5: Log

3.5 Apply to Site B

Open a fresh Copilot conversation. Work from your Word log — apply each saved prompt to Site B.

Save combined output as `plants_combined.csv`

Segment 4 — Wrap-up

4.1 Scale test

New Copilot conversation. Drag and drop `plants_sites_AB_large.csv`.

Prompt →

```
Here is a dataset of 1,520 plant survey records from two sites.
> What I want you to do is to complete all of the same steps you completed for
site_A and _B. Return a table of all original columns (even those not modified)
and `[column_name]_cleaned' for any new, cleaned ones.

> For numerical values, flag any potential outliers with "check"; for
categorical variables, change spelling, separation errors, and flag any values
you are uncertain about with "check".

> Perform these steps:

> Cleaned or new columns    Modification prompt
Height_cm   Look at `Height_cm` only. Return a two-column table with the
original column (`Hieght_cm`), and a new column (`Height_cm_cleaned`) with
potential outlier values marked with "check".
Num_Leaves
Num_Flowers Provide me the same summary stats and outlier checks for
`Num_Leaves` and `Num_Flowers`.

> Notes_themes  Provide me a row-by-row thematic analysis of the `Notes` field,
limit your analysis of each comment into on of three, short distinct themes from
the perspective of a botanist that you will decide. Include the analysis as
column `Notes_themes`.

> Site`Location`, `Quadrat`, `Species` and `Flowering`. Clean categorical
columns `Site`, `Location`, `Quadrat`, `Species` and `Flowering`.
I want them to be consistently spelt, all values lower-case, and values
separated by underscores "_".
Return pairs of columns for each variable: the original and a new column
`[column_name]_cleaned` for
each. In the cleaned column have the new, corrected value and use "check"" if
you are unsure.

> Using `Site_cleaned`, `Location_cleaned`, `Quadrat_cleaned`, and
`Species_cleaned`,
create a new column `Plant_ID`. Format example: Sa_L1_Q1_LS (Site = A, Location
= 1, quadrat = 1, Leptospermum scoparium)
Return the only the new column.

> Plant_Age_Months  Look at `Date_Planted` and `Date_Measured`. These columns
contain dates
in inconsistent formats — some DD/MM/YYYY, some D/M/YY, some DD-Mon
with no year, and possibly some US-format (M/D/YY).
Without standardising the formats first, calculate the age of each plant
in whole months at the time of measurement. Add this as a new column
called `Plant_Age_Months`. Return the these three columns as a table.
Flag any row where you cannot confidently interpret the dates with "check".
```

```
> Condition For `Condition`, return the original column and a new column
`Condition_cleaned` with suggested categories representing a consistent five-
point scale from poor to excellent.

> Data_Quality_Flag
Data_Quality_Notes  I am about to analyse 'Age_Months`, `Height_cm`, and
`Condition` for these
plants. For each row, I want to know whether the `Notes` field give any reason
this row's
measurements might be unreliable or need special handling.
Add two new columns: `Data_Quality_Flag`: Y or N; and `Data_Quality_Notes`: a
short reason for any Y."
```

4.2 AI R Code

Prompt →

```
Produce the R code to run these steps myself in one block of code (one script) I
can copy and paste across in one go. Put some thought into this.
```

4.3 Decision Matrix

4.4 Considerations

4.5 Resources + close